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CL-2 (IR) Assignment 4: Agglomerative Hierarchical Clustering

Implement the Agglomerative Hierarchical Clustering algorithm using an appropriate dataset.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from sklearn.metrics import silhouette_score
from scipy.cluster.hierarchy import dendrogram, linkage

plt.rcParams["figure.figsize"] = (10, 5)
```

```
# Use the Iris dataset as an appropriate numerical clustering dataset.
iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df["target"] = iris.target

display(df.head())
print("Dataset shape:", df.shape)
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
Dataset shape: (150, 5)
```

```
print("Columns:")
print(df.columns.tolist())

print("\nData types:")
display(df.dtypes.to_frame("dtype"))

print("\nMissing values:")
display(df.isna().sum().to_frame("missing"))

print("\nSummary statistics:")
display(df.describe().T)
```

```
Columns:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)', 'target']

Data types:
```

	dtype
sepal length (cm)	float64
sepal width (cm)	float64
petal length (cm)	float64
petal width (cm)	float64
target	int32

```
Missing values:
```

	missing
sepal length (cm)	0
sepal width (cm)	0
petal length (cm)	0
petal width (cm)	0
target	0

```
Summary statistics:
```

	count	mean	std	min	25%	50%	75%	max
sepal length (cm)	150.0	5.843333	0.828066	4.3	5.1	5.80	6.4	7.9
sepal width (cm)	150.0	3.057333	0.435866	2.0	2.8	3.00	3.3	4.4
petal length (cm)	150.0	3.758000	1.765298	1.0	1.6	4.35	5.1	6.9
petal width (cm)	150.0	1.199333	0.762238	0.1	0.3	1.30	1.8	2.5
target	150.0	1.000000	0.819232	0.0	0.0	1.00	2.0	2.0

```
X = df[iris.feature_names].copy()

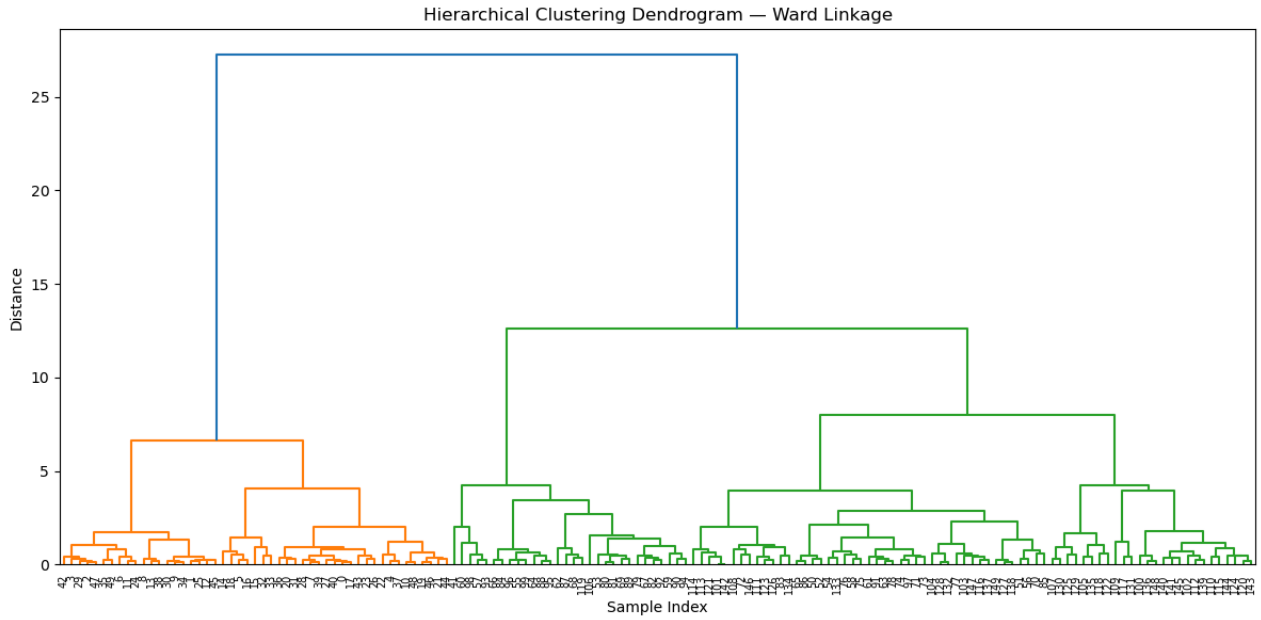
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_scaled_df = pd.DataFrame(X_scaled, columns=iris.feature_names)
display(X_scaled_df.head())
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	-0.900681	1.019004	-1.340227	-1.315444
1	-1.143017	-0.131979	-1.340227	-1.315444
2	-1.385353	0.328414	-1.397064	-1.315444
3	-1.506521	0.098217	-1.283389	-1.315444
4	-1.021849	1.249201	-1.340227	-1.315444

```
Z = linkage(X_scaled, method="ward")

plt.figure(figsize=(12, 6))
dendrogram(
    Z,
    labels=[str(i) for i in range(len(X_scaled))],
    leaf_rotation=90,
    leaf_font_size=7
)
plt.title("Hierarchical Clustering Dendrogram — Ward Linkage")
plt.xlabel("Sample Index")
plt.ylabel("Distance")
plt.tight_layout()
plt.show()
```




```
n_clusters = 3

model = AgglomerativeClustering(
    n_clusters=n_clusters,
    linkage="ward"
)

labels = model.fit_predict(X_scaled)
df["cluster"] = labels

display(df.head(10))
print("Cluster counts:")
display(df["cluster"].value_counts().sort_index())
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	cluster
0	5.1	3.5	1.4	0.2	0	1
1	4.9	3.0	1.4	0.2	0	1
2	4.7	3.2	1.3	0.2	0	1
3	4.6	3.1	1.5	0.2	0	1
4	5.0	3.6	1.4	0.2	0	1
5	5.4	3.9	1.7	0.4	0	1
6	4.6	3.4	1.4	0.3	0	1
7	5.0	3.4	1.5	0.2	0	1
8	4.4	2.9	1.4	0.2	0	1
9	4.9	3.1	1.5	0.1	0	1

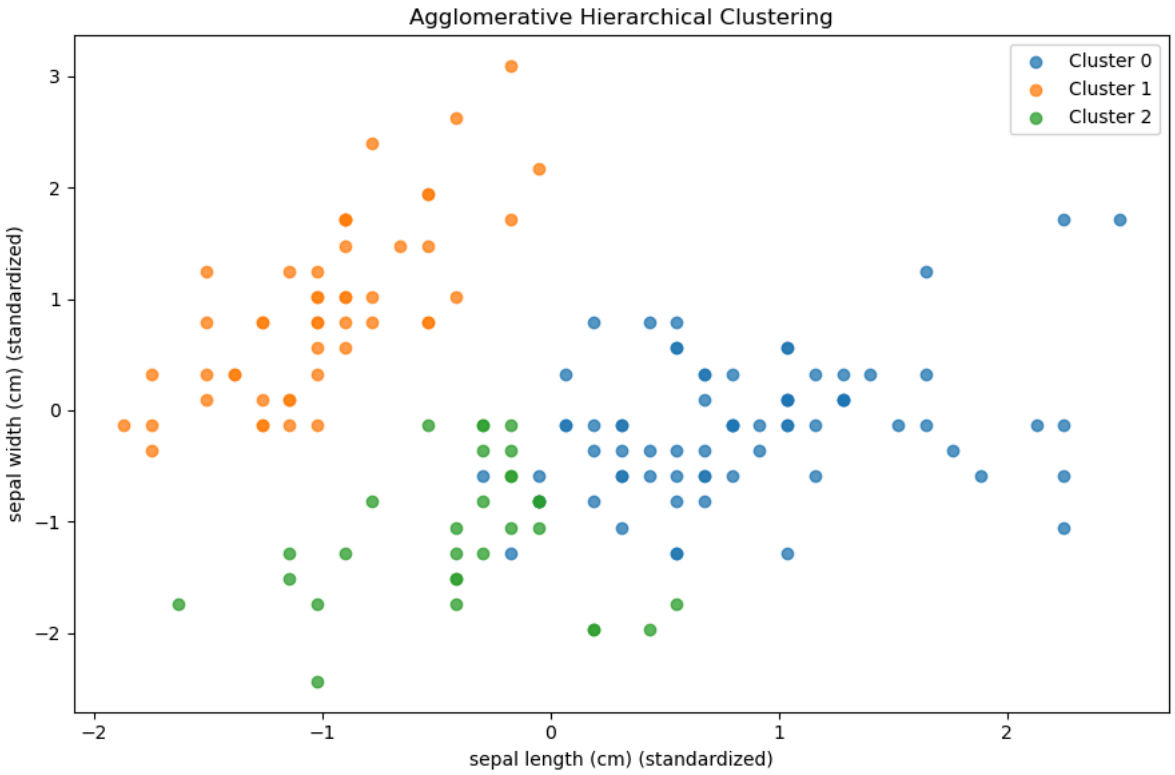
```
Cluster counts:
cluster
0    71
1    49
2    30
Name: count, dtype: int64
```

```
# Visualize using the first two standardized features.
plt.figure(figsize=(9, 6))

for cluster_id in sorted(df["cluster"].unique()):
    mask = df["cluster"] == cluster_id
    plt.scatter(
        X_scaled[mask, 0],
        X_scaled[mask, 1],
        label=f"Cluster {cluster_id}",
        alpha=0.75
```

```
    )

plt.title("Agglomerative Hierarchical Clustering")
plt.xlabel(iris.feature_names[0] + " (standardized)")
plt.ylabel(iris.feature_names[1] + " (standardized)")
plt.legend()
plt.tight_layout()
plt.show()
```




```
silhouette = silhouette_score(X_scaled, labels)
print(f"Silhouette Score: {silhouette:.4f}")
```

```
Silhouette Score: 0.4467
```